



Grade 11 Geography

5. Slopes

Learning Guide

 1. What is a Slope?

- A **slope** is the **angle that the Earth's surface makes with the horizontal**.
- It can also refer to any **geometric feature of the landscape**.
- Slopes are important because they show how land rises, falls, and changes shape over time.

 Key idea:

A slope tells us how steep or gentle a part of the land is.

 2. South Africa's Major Landform Regions

South Africa's topography includes several major landform regions:

1. The Plateau

- A relatively **flat interior** region.
- It is **lower in the west**, where the **Kalahari Basin** is found.
- It becomes **higher in the east**, where the land rises towards the escarpment.

2. The Escarpment

- A mountainous boundary that separates the **plateau** from the **coastal plain**.
- It consists of a number of mountain ranges.
- The **Drakensberg** in the east forms the highest part of the escarpment.

3. Slopes of the Escarpment

- These are the slopes where land drops from the **mountain tops** down towards the **coastal plain**.

4. The Coastal Plain

- A low-lying strip of land found around South Africa's edges.
- It has a generally **low altitude**, except where the **Cape Fold Mountains** occur.

5. Cape Fold Mountains

- Steep and rugged mountains.
- Two bands extend **west to east** in the southern coastal plain.
- One band extends **north to south** in the west.
- These mountain ranges meet at the **Hex River Mountains**.

 Diagram note:

The **west-to-east cross-section** shows the coastal plain, plateau, escarpment, and slopes of the escarpment.

 3. Types of Slopes

1. Gentle Slope

- A **gentle slope** has a gradual rise or fall.
- On a map, it is shown by **widely spaced contour lines**.
- The slope is usually even and not difficult to walk on.
- Common in:
 - **Flood plains**
 - **Plateaus**

 Mapwork tip:

Wide contour spacing = gentle slope.

2. Steep Slope

- A **steep slope** rises or falls suddenly.
- On a map, it is shown by **closely spaced contour lines**.
- Often found near:
 - **Cliffs**
 - **Escarpments**

📌 **Mapwork tip:**

Close contour spacing = steep slope.

3. Convex Slope

- A **convex slope** becomes steeper uphill.
- The upper part is steeper than the lower part.
- On a map, the contour lines become **closer together towards the top**.
- It may be seen where erosion increases with height.

📌 **Key idea:** Convex slopes curve outward and steepen towards the top.

4. Concave Slope

- A **concave slope** also steepens uphill.
- It has **widely spaced contours at the base** and **closer contours at the top**.
- Often found in **humid regions**.
- The lower part is usually gentler, while the upper part becomes steeper.

📌 **Key idea:** Concave slopes curve inward and often become gentler towards the base.

4. Slopes Formed by Tectonic Forces: Primary Slopes

- Some slopes are formed by **tectonic forces**.
- These forces come from inside the Earth and are linked to:
 - **Folding**
 - **Faulting**

Examples

- **Anticline slopes**
- **Syncline slopes**
- **Fault scarp**

Fault Scarp

- A fault scarp forms where land moves **vertically** along a fault.
- One block of land may be raised while another is lowered.

📌 **Key idea:** Primary slopes are formed by internal Earth forces such as folding and faulting.

5. Slopes Formed by Erosion

1. Primary Erosion Slope

- A primary erosion slope is created by **one main agent of erosion**.
- Example:
 - A river cutting down into a plateau creates **canyon walls**.
- Water erodes **vertically downward**, forming steep valley sides.

📌 **Key idea:** One erosion agent shapes the slope.

2. Secondary Erosion Slope

- A secondary erosion slope is further shaped by **running water or weathering**.
- It forms through **more than one form of erosion**.
- Example:
 - A river deepens a valley while water also erodes the sides.

✚ **Key idea:** Secondary erosion slopes are modified by several erosion processes.

6. Slopes Formed by Deposition

- Depositional slopes form when material is **deposited** or laid down.
- These slopes are built up by accumulated material.

Examples

- **Sand dunes**
- **River alluvial fans**
- **Foot slopes with weathered debris**

✚ **Key idea:** Deposition builds slopes, while erosion wears slopes down.

7. Main Slope Elements

A slope is made up of different parts called **slope elements**.

1. Crest

- The **top of the slope**.
- Usually **convex** in shape.
- Often has **thin soil**.

✚ **Key idea:** Crest = uppermost part of a slope.

2. Cliff / Scarp

- A **steep rock face**.
- Usually made of **hard rock**.
- Rockfall often occurs here.
- It may retreat backwards over time.

✚ **Key idea:**

Cliff/scarp = steep, exposed rock face.

3. Talus

- A slope made of rock debris from the cliff.
- Also called **scree**.
- Forms when rocks fall and collect at the base of the cliff.

✚ **Key idea:** Talus = pile of broken rock below the cliff.

4. Pediment

- A **gentle, concave slope** at the base.
- It grows as the slope retreats.
- Usually found below the talus slope.

✚ **Key idea:**

Pediment = gentle lower slope.

5. Knickpoint

- A **knickpoint** marks a change in gradient between two slope elements.
- It is usually found between the **talus** and **pediment**.

✚ **Key idea:**

Knickpoint = sudden change in slope angle.

8. Characteristics of Slope Elements

Slope Element	Key Characteristics
Crest	Convex, thin soil, top of hill
Cliff / Scarp	Vertical, hard rock, parallel retreat
Talus	Scree/debris slope, formed by rockfall
Pediment	Gentle, concave, grows as slope retreats

9. Slope Development Over Time

- Slopes change due to:
 - **Weathering**
 - **Erosion**
 - **Mass movement**

General Process

1. The **cliff** is weathered and eroded.
2. Rock breaks off and collects below as **talus**.
3. The talus slope grows.
4. The lower slope becomes more **concave**.
5. The cliff or scarp gradually moves backward.

10. Slope Retreat / Scarp Retreat

- **Slope retreat** happens when a slope, especially a cliff or scarp, moves **backwards over time**.
- The slope is worn away by:
 - Weathering
 - Erosion
 - Rockfalls
 - Mass movement

What happens?

- The cliff face is eroded.
- Rock debris falls and forms a talus slope.
- The pediment grows at the base.
- Over time, the steep part of the slope shifts backwards.

Key idea:

Slope retreat means the slope moves backwards while the landscape becomes wider and more worn down.

11. Final Exam Summary

- A **slope** is the angle the Earth's surface makes with the horizontal.
- South Africa's main landform regions include:
 - Plateau
 - Escarpment
 - Coastal plain
 - Cape Fold Mountains
- **Gentle slopes** have widely spaced contours.
- **Steep slopes** have closely spaced contours.
- **Convex slopes** steepen uphill.
- **Concave slopes** are often found in humid regions and have wider spacing at the base.
- **Primary slopes** form through tectonic forces such as folding and faulting.
- **Erosion slopes** form through river action, weathering, and running water.
- **Depositional slopes** form where material is deposited.
- Main slope elements:
 - Crest, Cliff/scarp, Talus, Pediment, Knickpoint
- Slopes develop over time through **weathering, erosion, and mass movement**.
- **Scarp retreat** occurs when a cliff face moves backward over time.

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